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Review Article

A comparative analytical study of LLM-based automated grading in programming education

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A B S T R A C T

The integration of Large Language Models (LLMs) into programming education has opened new avenues for automated assessment, extending beyond traditional test-case-based evaluation. Despite the rapid proliferation of LLM-based grading research, the literature remains methodologically fragmented, making it difficult to establish robust evaluation practices and emerging design principles. Existing reviews provide broad conceptual perspectives on AI in education and automated assessment, but few offer a grading-specific, taxonomy-driven analytical framework. As research in this area rapidly expands across heterogeneous educational contexts, a structured comparative synthesis is both feasible and increasingly necessary.

This paper presents a comparative analytical study of LLM-based automated grading systems in computer science education. Following a PRISMA-guided methodology, we systematically analyze a corpus of 17 peer-reviewed studies (2023–2026), including high-impact journals and leading computing education conferences. We develop an inductively derived taxonomy that characterizes existing systems across five dimensions: grading paradigm, role of the LLM, human involvement, evaluation strategy, and generalization scope. The reviewed literature is explicitly stratified by level of evidence to distinguish exploratory pilots from more mature empirical evaluations.

Our analysis indicates that fully automated direct-grading approaches still dominate the current literature, while interest in hybrid and human-supervised configurations—in which LLMs increasingly function as triaging or assistance mechanisms—is clearly growing but not yet prevalent. Current studies rely predominantly on agreement-based evaluation metrics, particularly correlation-based comparisons with human graders, whereas cross-rubric validation and pedagogically grounded assessment measures remain limited. We adopt a structured review approach because the principal challenge in this emerging field lies not only in generating additional experiments, but in establishing a coherent framework through which heterogeneous findings can be interpreted and compared. Overall, this review structures the current exploratory landscape and offers actionable insights toward more robust, transparent, and educationally grounded AI-assisted grading frameworks.

1. Introduction

The increasing scale of programming education, particularly in higher education and professional training contexts, has intensified long-standing challenges related to assessment. Manual grading of programming assignments is time-consuming, difficult to scale, and often affected by subjectivity and inconsistency across instructors (Cheang et al., 2003) (Gao et al., 2024). While traditional automated grading systems based on test cases and static analysis offer efficiency, they frequently fall short in capturing partial correctness, alternative solution strategies, and pedagogical intent—especially for novice programmers (Zampirolli et al., 2026).

Recent advances in large language models (LLMs) have introduced new possibilities for automated assessment in programming education (Moorhouse et al., 2023). Unlike rule-based autograders, LLMs are capable of interpreting source code in a more holistic manner, reasoning about program logic, structure, and intent (Zampirolli et al., 2026). As a result, an increasing number of studies have explored the use of LLMs for grading programming tasks, reporting promising levels of agreement with human instructors and significant reductions in grading effort (Jukiewicz, 2024).

In this study, automated grading refers to the use of computational systems to assign scores, categories, or evaluative judgments to programming submissions with minimal or no direct human intervention at the time of grading.

However, despite this rapid growth, the current body of research remains fragmented (Moorhouse et al., 2023). Studies differ substantially in how automated grading is conceptualized, how LLMs are integrated into assessment workflows, and how performance is evaluated. Some approaches focus on binary correctness or category-based grading, while others aim to predict continuous scores that approximate instructor-assigned grades. Similarly, the role of human instructors ranges from full automation to tightly coupled human-in-the-loop configurations. These methodological differences complicate the

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interpretation and comparison of reported results across studies.

This fragmentation is not merely a descriptive observation but a methodological challenge that directly affects the interpretability of reported results. When studies differ in grading paradigms, evaluation metrics, and levels of human involvement, comparisons of performance become inherently unstable. Consequently, similar reported outcomes may reflect fundamentally different assessment assumptions rather than comparable grading capabilities. This creates a need for structured comparative analysis that focuses explicitly on methodological design rather than performance alone.

For example, some studies evaluate grading performance using exact-match accuracy in constrained tasks, while others rely on correlation or mean absolute error to approximate instructor-assigned scores. In parallel, certain approaches adopt fully automated grading configurations, whereas others incorporate human-in-the-loop validation, leading to fundamentally different interpretations of what constitutes effective grading.

Moreover, evaluation practices vary widely. Most existing work relies on agreement-based metrics that quantify numerical similarity between AI-generated and human-assigned grades (Jukiewicz, 2024) (Messer et al., 2024), implicitly treating instructor grading as a stable and unproblematic ground truth (Jukiewicz, 2024). Fewer studies examine statistical equivalence or address the pedagogical implications of grading discrepancies. In addition, evaluations are typically conducted within narrow educational contexts, such as a single course, exam, or programming language, raising questions about the generalizability of current findings (Messer et al., 2024) (Zampirolli et al., 2026).

While prior review studies have examined automated assessment and feedback tools in programming education (e.g., Messer et al. (Messer et al., 2024)), their primary focus lies in the broader ecosystem of programming assessment technologies, including unit testing, static analysis, code-quality evaluation, language paradigms, and automation levels. By contrast, the methodological diversity introduced specifically by LLM-based grading systems—including grading paradigms, human involvement configurations, evaluation assumptions, and evidence heterogeneity—remains comparatively underexamined.

As a result, the current challenge in this rapidly evolving area lies less in the absence of individual empirical studies than in the lack of a coherent methodological framework for interpreting and comparing them. Existing findings are often reported using incompatible methodological assumptions, making it difficult to distinguish between differences arising from grading design choices, evaluation protocols, or contextual constraints.

This need becomes particularly urgent as LLM-based grading research rapidly expands across heterogeneous educational contexts and publication venues.

Accordingly, this paper presents a comparative analytical study of recent research on LLM-based automated grading in programming education. Drawing on an expanded corpus of 17 peer-reviewed studies (2023–2026), including leading journals and computing education conferences, the paper proposes a taxonomy that captures key methodological dimensions of existing approaches. Building on this taxonomy, the study systematically compares evaluation practices, implicit assumptions, and methodological limitations across the reviewed corpus.

This study makes the following contributions:

- A taxonomy-driven framework for analyzing LLM-based automated grading approaches is proposed.
- Key methodological dimensions shaping current research are identified and synthesized.
- Implicit assumptions underlying evaluation practices are critically examined.
- An evidence-aware comparative perspective is introduced to differentiate exploratory studies, case-based evaluations, and more mature empirical investigations.
- Systematic methodological gaps limiting interpretability, validation, and generalizability are highlighted.

From a practical standpoint, the findings of this analysis carry direct implications for educators and institutions. Understanding how different grading paradigms, levels of automation, and evaluation strategies influence assessment outcomes is essential for making informed decisions about adopting LLM-based grading tools. For example, fully automated approaches may offer scalability for large enrollment courses with well-defined tasks, while human-in-the-loop configurations may better serve contexts where pedagogical nuance and accountability are priorities. By clarifying these trade-offs, this study aims to support more evidence-based adoption of automated grading technologies in programming education.

Rather than benchmarking specific models or advocating a single optimal solution, this work aims to clarify the methodological landscape of LLM-based automated grading. By identifying common patterns and persistent gaps, the paper seeks to support more coherent, pedagogically grounded, and generalizable research in this emerging area of AI-enhanced programming education.

2. Related work

Research on automated assessment in programming education has evolved over several decades, initially relying on rule-based autograders, test-case execution, and static analysis techniques. While these approaches offer scalability and objectivity, they are often limited in their ability to evaluate partial correctness, alternative solution strategies, and higher-order reasoning. Consequently, recent advances in large language models (LLMs) have stimulated growing interest in rethinking automated grading practices.

The evolution of automated grading in programming education can be broadly organized into four overlapping stages, each reflecting shifts in technology, educational scale, and assessment philosophy.

(1) Rule-based and test-case evaluation (1990s–2000s).

Early automated grading systems primarily relied on compiling and executing student submissions against predefined test suites (Ala-Mutka, 2005) (Douce et al., 2005) (Hollingsworth, 1960). Foundational reviews by Ala-Mutka (Ala-Mutka, 2005) and Douce et al. (Douce et al., 2005) traced the evolution of these systems from early assessment tools introduced by Hollingsworth (Hollingsworth, 1960) toward increasingly structured web-based grading environments. Rule-based approaches offered objective and scalable evaluation for well-defined programming tasks but largely focused on output correctness, with limited support for assessing partial credit, code quality, or alternative solution strategies.

(2) Scalable assessment, static analysis, and online learning infrastructures (2000s–2010s).

The expansion of programming education—particularly through large-scale online courses and MOOCs—created demand for highly scalable assessment mechanisms (Paiva et al., 2022), (Souza et al., 2016). During this period, platforms such as Web-CAT and AutoGradr popularized unit-test-based grading workflows capable of providing rapid feedback and supporting repeated submissions (Paiva et al., 2022). In parallel, static analysis techniques became increasingly integrated into automated assessment, including linters, software metrics, style checking, and code-similarity approaches (Ala-Mutka, 2005), (Paiva et al., 2022). Research also established plagiarism detection, comparative code analysis, and hybrid dynamic–static approaches as distinct areas concerned with academic integrity, structural similarity, and

broader assessment flexibility (Ala-Mutka, 2005), (Ullah et al., 2018). Although these approaches broadened automated assessment beyond simple execution testing, they remained constrained in their ability to interpret semantic intent, pedagogical nuance, or open-ended solution design (Ala-Mutka, 2005) (Souza et al., 2016).

(3) Machine learning and neural code models (2010s–early 2020s).

The rise of machine learning introduced more flexible assessment paradigms based on learned representations rather than manually specified rules (Ala-Mutka, 2005). Following the extension of automated assessment research toward ML-based AATs, researchers explored neural and feature-driven models for grading correctness, design quality, feedback generation, and knowledge tracing (Dong et al., 2020), (Efremov et al.), (Orr & Russell, 2021). These approaches frequently depended on large training datasets, engineered features, or task-specific supervision (Dong et al., 2020) (Efremov et al.). Later developments in transformer-based code representation—including models such as CodeBERT (Feng et al., 2020) and GraphCodeBERT (Guo et al., 2021) —enabled richer semantic modelling for grading-adjacent tasks such as code understanding, summarization, and vulnerability detection. However, these systems generally required fine-tuning and lacked the generative flexibility necessary for nuanced, rubric-aware evaluation (Dong et al., 2020), (Efremov et al.).

(4) LLM-based grading systems (2023–present).

The emergence of large language models represents the most recent paradigm shift in automated grading (Pwanedo Amos et al., 2025). Unlike earlier approaches, LLMs are capable of interpreting source code semantically, reasoning about program logic and intent, and generating context-aware evaluations without extensive task-specific engineering (Jiang & Fowler, 2026), (Bernik, Radošević, & Čep, 2025). This capability has expanded automated assessment beyond binary correctness checking toward continuous score prediction, rubric-based grading, explanatory feedback generation, and human–AI collaborative workflows (Mendonça et al., 2025), (Bernik, Radošević, & Čep, 2025).

This historical trajectory reflects a broader transition from syntactic and structural evaluation toward semantic, context-aware, and increasingly pedagogically informed assessment paradigms (Pwanedo Amos et al., 2025). At the same time, it highlights a persistent tension between scalability, interpretability, pedagogical richness, and validation rigor—an issue that becomes particularly salient in contemporary LLM-based grading research (Jiang & Fowler, 2026), (Mendonça et al., 2025), (Bernik, Radošević, & Čep, 2025).

2.1. LLMs for automated grading in programming education

Recent studies have explored the use of LLMs to assess programming assignments by directly generating grades, categories, or feedback based on source code submissions. These approaches differ substantially in how grading is operationalized. Some studies frame automated grading as a **binary or category-based decision**, particularly in constrained domains such as SQL or introductory exercises (Manikani et al., 2025), (Yousef et al., 2025), whereas others aim to **predict continuous scores** that approximate instructor-assigned grades (Mendonça et al., 2025), (Mahdaoui et al., 2025). A smaller subset of work adopts **rubric-based analytical grading** (Bernik, Radošević, & Čep, 2025), Bernik, Radosevic, & Cep, 2025ecomposing assessment into multiple criteria aligned with pedagogical objectives.

This diversity reflects the exploratory stage of the field but also complicates cross-study comparison, as reported performance metrics depend heavily on how grading itself is defined.

2.2. Human involvement in LLM-based grading systems

Another line of differentiation concerns the role of human instructors in LLM-based grading workflows. Some studies pursue fully automated configurations, positioning LLMs as direct graders (Jukiewicz, 2024) whose outputs are evaluated post hoc against human assessments. In contrast, pedagogically grounded approaches often retain instructors in the grading loop, using LLMs as assistants that support, rather than replace, human judgment (Bernik, Radošević, & Čep, 2025).

These differing configurations suggest that human involvement is not merely a transitional limitation (Bernik, Radošević, & Čep, 2025) (Mendonça et al., 2025) but a deliberate design choice shaped by educational context, accountability requirements, and assessment goals.

2.3. Evaluation practices and validation strategies

Evaluation methodologies in LLM-based automated grading studies are similarly heterogeneous. Most existing work relies on agreement-based metrics (Mendonça et al., 2025) (Jukiewicz, 2024), such as correlation coefficients or error measures, to quantify similarity between AI-generated and human-assigned grades. A smaller number of studies employ accuracy-based metrics (Manikani et al., 2025), particularly when grading outcomes are discrete or binary. Only limited work explores statistical equivalence testing (Mendonça et al., 2025), which provides a more stringent validation of grading interchangeability.

While these evaluation strategies offer insights into model behavior, they also embed implicit assumptions about the reliability and pedagogical adequacy of human grading, which are rarely examined explicitly.

2.4. Limitations of existing reviews

Although several survey and review studies examine automated assessment, feedback systems, and broader applications of artificial intelligence and large language models in programming education (Messer et al., 2024) (Pwanedo Amos et al., 2025) (Pereira & Ferreira Mello, 2025), few focus specifically on LLM-based automated grading in programming education, and even fewer adopt a comparative, methodology-oriented perspective. Existing reviews often emphasize technical capabilities or reported performance gains without systematically analyzing how design choices influence evaluation outcomes and generalizability (Weißenrieder et al., 2025).

2.5. Positioning of the present study

Building on this body of work, the present study differs from prior reviews by concentrating explicitly on methodological diversity rather than model performance. By synthesizing recent LLM-based grading studies through a taxonomy-driven comparative analysis, this work aims to clarify how grading paradigms, human involvement, and evaluation strategies shape current findings and to identify persistent gaps that currently limit the interpretability and broader applicability of the field.

3. Methodology

3.1. Study design

This study adopts a comparative analytical research design aimed at systematically examining methodological patterns in recent LLM-based automated grading research within programming education. Rather than empirically benchmarking model performance, the study focuses on how existing work conceptualizes grading, integrates large language models, incorporates human involvement, and validates assessment outcomes.

3.2. Search strategy and database selection

The literature search was conducted using Scopus and Web of Science (WoS). These databases were selected because they provide broad, curated coverage of peer-reviewed computer science, artificial intelligence, and computing education research, and because they index a substantial portion of the major journal and conference literature published through ACM and IEEE, the principal publication outlets in this field. Their consistent metadata also facilitates reproducible query construction and study retrieval across databases. Nevertheless, these databases do not index all relevant venues exhaustively, and some specialized computing education proceedings may have incomplete coverage. This limitation is discussed further in Section 8.

To ensure broad coverage of the research area, three complementary search strategies were employed.

3.2.1. Query A: Core LLM-Based automated grading studies

("automated grading" OR "automated assessment" OR "automated scoring" OR "automated evaluation" OR "auto-grading" OR "autograding") AND ("large language model" OR "LLM" OR "GPT" OR "ChatGPT" OR "GPT-4″ OR "Gemini" OR "Claude" OR "Llama" OR "generative AI") AND ("programming" OR "code" OR "coding" OR "computer science education" OR "programming education" OR "programming assignments")

3.2.2. Query B: expanded search strategy

("LLM" OR "large language model" OR "GPT-4″ OR "generative AI") AND ("grading" OR "scoring" OR "assessment" OR "evaluation") AND ("programming" OR "code" OR "student submissions" OR "student code")

3.2.3. Query C: Feedback–Grading intersection

("feedback" OR "automated feedback") AND ("grading" OR "scoring" OR "grade" OR "score") AND ("LLM" OR "large language model" OR "GPT") AND ("programming" OR "code")

Equivalent search logic was implemented in both databases using their respective search syntaxes and title/abstract/keyword fields.

The search scope was restricted to studies published between **2023 and 2026** to capture both early exploratory applications and more recent methodological developments in LLM-based grading research.

Retrieved records were subsequently filtered according to predefined criteria:

- **Document type:** Article, Conference Paper, Review.

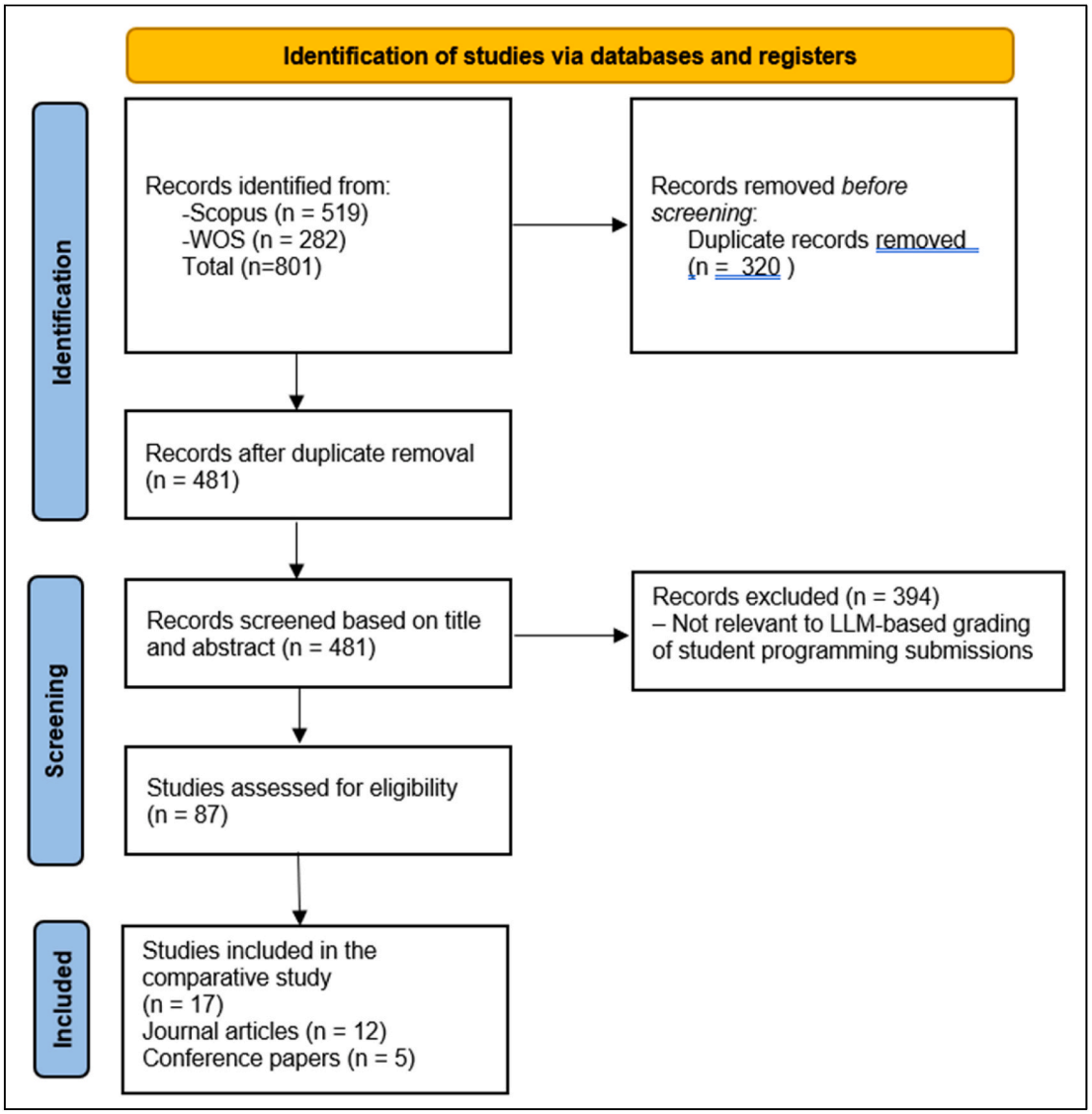


Fig. 1. PRISMA-inspired flow diagram illustrating the study identification, screening, and selection process.

- **Research area/subject area:** Computer Science, Engineering, Education, and related interdisciplinary domains.
- **Language:** English.

The inclusion of both journal articles and conference proceedings reflects disciplinary publication practices in computer science, where conferences frequently constitute major outlets for high-impact peer-reviewed research.

3.3. Study selection and screening procedure

As summarized in Fig. 1, database searches identified 801 records (Scopus, n = 519; Web of Science, n = 282). After removal of 320 duplicates, 481 unique records remained and were screened by title and abstract, of which 394 were excluded as not relevant to LLM-based grading of student programming submissions. The remaining 87 records were assessed for eligibility through full-text review, yielding a final corpus of 17 studies (12 journal articles and 5 conference papers).

Following database retrieval, records were organized, deduplicated, and screened using predefined inclusion and exclusion criteria applied to titles, abstracts, and full texts.

Studies were included if they satisfied all of the following conditions:

(a) explicit use of one or more large language models for grading, scoring, or evaluative assessment of student-written programming code;
(b) application within a programming education context, including assignments, examinations, coursework, laboratory activities, or related instructional settings;
(c) publication in a peer-reviewed journal article, conference proceeding, or review venue;
(d) availability of sufficient methodological detail to support comparative analysis.

Studies were excluded if they:

(a) focused exclusively on feedback generation, natural-language explanation assessment, or essay grading without an explicit grading or scoring component;
(b) evaluated code generation capabilities rather than the assessment of student-written code;
(c) consisted of editorials, workshop abstracts, preprints, or publications lacking sufficient methodological detail.

Two of the three authors independently screened the titles and abstracts of the deduplicated records against the inclusion and exclusion criteria, retaining studies coherent with the aim of the review. Full texts of the retained records were then assessed for eligibility. The third author independently reviewed and validated the resulting selection, and any disagreements were resolved through discussion until consensus was reached. Inter-rater agreement was assessed through independent screening and consensus resolution; however, no formal agreement coefficient was calculated.

The resulting corpus consisted of **17 peer-reviewed studies**, spanning **2023–2026** and including both journal publications and major computing education conference venues.

3.4. Analytical framework and taxonomy development

A structured analytical framework was developed to guide the comparative analysis.

The taxonomy was constructed inductively through an iterative multi-pass coding process applied to the selected studies.

Step 1 — Open Coding:

Each study was examined in full, and methodological attributes related to grading design, LLM integration, human involvement, evaluation practices, and validation scope were annotated.

Step 2 — Axial Coding:

The extracted attributes were grouped into thematic clusters to identify recurring patterns, methodological similarities, and contrasts across studies.

Step 3 — Selective Coding and Consolidation:

The resulting thematic clusters were refined into five final taxonomy dimensions:

1. **Grading Paradigm** — binary/test-oriented, discrete category-based, continuous score prediction, or rubric-based analytical grading.
2. **Role of the LLM** — direct grader, grading assistant, hybrid architecture, or fine-tuned grading system.
3. **Human Involvement** — fully automated, human-on-the-loop, or human-in-the-loop configurations.
4. **Evaluation Strategy** — agreement-based, accuracy-based, or equivalence-testing approaches (correlation, MAE, R^2, absolute difference, and classification accuracy against human labels are treated as agreement-based sub-metrics).
5. **Generalization Scope** — ranging from single-course evaluations to broader multi-context or comparative validation settings.

These dimensions serve both as a classification framework and as an analytical lens for identifying recurring assumptions, methodological tensions, and systematic gaps across the reviewed literature.

3.5. Level of evidence classification

Given the methodological heterogeneity of the reviewed corpus, each study was additionally classified according to its **level of evidence**.

The classification distinguishes:

- **Exploratory pilots:** feasibility-oriented studies with limited scope or informal validation.
- **Case studies:** context-specific investigations conducted within a single course, institution, or instructional setting.
- **Controlled experimental studies:** evaluations employing explicit comparison protocols and quantitative validation procedures.
- **Large-scale or comparative experimental studies:** broader investigations involving multiple datasets, institutions, educational settings, or systematic model comparisons.

Classification decisions were based on study scale, validation design, evaluation rigor, and reporting depth.

This stratification enables more nuanced interpretation of reported findings by accounting for differences in evidential strength rather than treating all included studies as methodologically equivalent.

3.6. Data extraction and synthesis

For each included study, methodological attributes were systematically extracted and documented within a structured comparison matrix.

The extraction process captured:

- bibliographic information;
- grading paradigm and LLM integration characteristics;
- human involvement configuration;
- evaluation strategy and validation metrics;
- generalization scope and reported limitations;
- implicit methodological assumptions;

- level-of-evidence classification.

The extracted information was synthesized using complementary comparative tables supported by narrative analytical discussion. This synthesis enabled the identification of recurring methodological patterns, validation practices, implicit assumptions, and persistent gaps across contemporary LLM-based automated grading research.

3.7. Scope and purpose

The objective of this methodology is not to rank or benchmark individual models but to clarify the methodological landscape of LLM-based automated grading research in programming education. By emphasizing design choices, validation practices, and evidence structures rather than isolated performance outcomes, the study aims to support more coherent interpretation of existing findings and inform future research directions.

4. Taxonomy of LLM-based automated grading approaches

To enable a systematic and coherent comparison of existing work, this study proposes a taxonomy that captures the key methodological dimensions of LLM-based automated grading in programming education. The taxonomy was derived iteratively through close reading of the selected studies, with attention to recurring design choices that directly influence how grading is conceptualized, implemented, and evaluated.

Specifically, the taxonomy characterizes each study along five dimensions: grading paradigm, role of the LLM, degree of human involvement, evaluation strategy, and generalization scope (see Fig. 2). These dimensions were selected because they consistently emerged as sources of variation across the reviewed corpus and were found to shape both reported performance and the interpretation of results. Rather than focusing on specific model architectures or numerical outcomes, the taxonomy emphasizes methodological structure and assessment design, which are central to understanding differences between studies.

Tables 1 and 2 operationalize this taxonomy by mapping each reviewed study to the identified dimensions and by synthesizing evaluation focus, implicit assumptions, and methodological limitations. Together, the taxonomy and comparative tables provide an analytical framework that supports interpretation of methodological diversity in current LLM-based automated grading research and serves as a foundation for the subsequent comparative discussion.

4.1. Taxonomy coding scheme

To improve clarity and readability, the taxonomy adopts a compact coding scheme to represent recurring grading configurations. *Grading paradigms* are categorized into four types: **Type I** denotes binary or correctness-based grading, **Type II** refers to discrete category-based grading, **Type III** represents continuous score prediction, and **Type IV** corresponds to rubric-based analytical grading. Similarly, human involvement is categorized into three levels:

Level 0 (*Fully Automated*), **Level 1** (*Human-on-the-Loop*), and **Level 2** (*Human-in-the-Loop*). Level 1 configurations allow selective instructor oversight or intervention, whereas Level 2 configurations involve active instructor participation in the grading process.

Evaluation strategies are grouped into three categories, applied consistently in Tables 1 and 3. Agreement-based evaluation is an umbrella category covering every metric that quantifies proximity to human-assigned grades — correlation, MAE, R^2, absolute difference, or classification accuracy relative to human labels — with the specific metric shown in parentheses in Table 1. Accuracy-based evaluation is reserved for studies that assess correctness against a predefined reference or expected output (e.g., test-oriented or benchmark grading), as distinct from accuracy computed against human labels. Equivalence testing denotes formal statistical equivalence procedures (e.g., TOST). This distinction explains why accuracy appears both as an agreement-based sub-metric and as a separate top-level category in Tables 1 and 3

As shown in Table 3, the reviewed literature is characterized by a strong methodological concentration around continuous score prediction, fully automated grading configurations, and agreement-based validation strategies. This pattern suggests that current research largely prioritizes numerical alignment with human grading over broader questions of pedagogical validity, cross-context robustness, or long-term educational impact. Although rubric-based and human-supervised approaches are increasingly represented, they remain less prevalent and are typically confined to narrower instructional contexts. The relative scarcity of equivalence testing and large-scale human-centered evaluation highlights the exploratory nature of the field and indicates important directions for future methodological development.

5. Comparative discussion

This comparative analysis reveals that current research on LLM-based automated grading in programming education is characterized less by disagreement in outcomes than by fundamental methodological diversity. The central argument developed across this section is that the field's dominant design choices are driven primarily by what is easy to measure and operationalize, rather than by what is pedagogically most informative—and that this measurement-driven orientation systematically shapes, and limits, the conclusions the field can currently support.

5.1. Continuous score prediction dominates because it is measurable, not because it is pedagogically optimal

The predominance of continuous score prediction (adopted by nearly

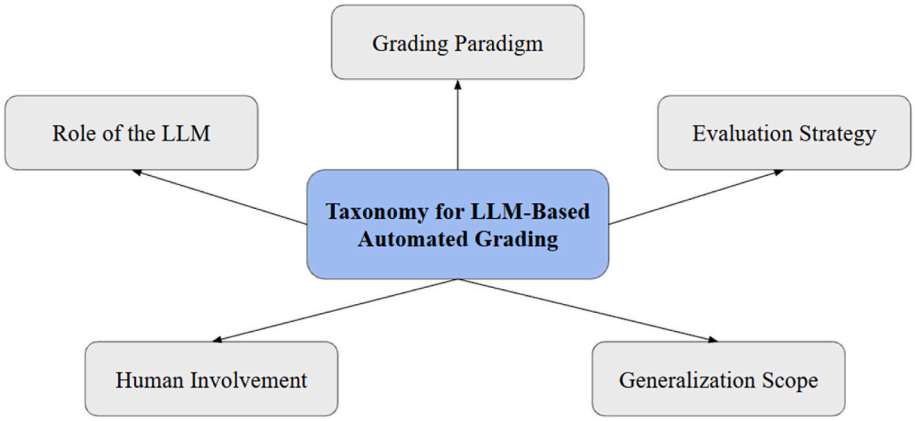


Fig. 2. Conceptual overview of the proposed taxonomy for LLM-based automated grading approaches in programming education.

Table 1
Comparative taxonomy of LLM-Based automated grading studies.

Study	Level of Evidence	Grading Paradigm	Role of the LLM	Human Involvement	Evaluation Strategy	Generalization Scope
Cisneros-González et al. (2025)	Case Study	Type IV: Rubric-Based Analytical Grading	Assistant to Human Grader	Level 2: Human-in-the-Loop	Agreement-based (MAE, R^2)	Single-Course/Single-Exam Validation
Mohamed et al. (2025)	Exploratory Pilot	Type III: Continuous Score Prediction	Direct Grader (Prompt-Based)	Level 0: Fully Automated	Accuracy-Based Evaluation	Limited-Task Validation
Mahdaoui et al. (2025)	Controlled Experimental Study	Type III: Continuous Score Prediction	Hybrid (LLM + ML)	Level 1: Human-on-the-Loop	Agreement-Based (MAE, R^2)	Single-Language Validation (Python)
Yousef et al. (2025)	Controlled Experimental Study	Type II: Discrete Category-Based Grading	Fine-Tuned Grader	Level 0: Fully Automated	Agreement-Based (Absolute Difference)	Single-Dataset Validation
Mendonça et al. (2025)	Controlled Experimental Study	Type III: Continuous Score Prediction	Direct Grader	Level 0: Fully Automated	Equivalence Testing	Single-Exam Validation
Karsa and Goldschmidt (2025)	Case Study	Type IV: Rubric-Based Analytical Grading	Direct Grader (Prompt-Based)	Level 2: Human-in-the-Loop	Agreement-Based (Correlation)	Multi-Exam Validation (Same Course)
Bernik, Radosevic, and Cep (2025)	Exploratory Pilot	Type III: Continuous Score Prediction (Preliminary)	Direct Grader	Level 2: Human-in-the-Loop	Agreement-Based (Correlation)	Single-Course Validation
Manikani et al. (2025)	Controlled Experimental Study	Type I: Binary/Test-Oriented Grading	Fine-Tuned Grader	Level 0: Fully Automated	Accuracy-Based Evaluation	Domain-Specific Validation (SQL)
Jukiewicz (2026)	Large-Scale Experimental Study	Type III: Continuous Score Prediction	Direct Grader	Level 0: Fully Automated	Agreement-Based (Correlation)	Multi-Year Single-Course Validation
Dosaru et al. (2025)	Case Study	Type II: Discrete Category-Based Grading	Direct Grader	Level 2: Human-in-the-Loop	Agreement-Based (Accuracy)	Single-Course Validation
Korthals et al. (2026)	Controlled Experimental Study	Type III: Continuous Score Prediction	Hybrid (LLM + ML)	Level 1: Human-on-the-Loop	Agreement-Based (Accuracy)	Multi-Exam Validation
Ibáñez et al. (2025)	Controlled Experimental Study	Type III: Continuous Score Prediction	Direct Grader	Level 0: Fully Automated	Agreement-Based (Correlation)	Single-Course Validation
Mazzone et al. (2025)	Controlled Experimental Study	Type IV: Rubric-Based Analytical Grading	Direct Grader	Level 0: Fully Automated	Agreement-Based (Accuracy)	Single-Course Validation
Pathak et al. (2025)	Controlled Experimental Study	Type IV: Rubric-Based Analytical Grading	Direct Grader	Level 0: Fully Automated	Agreement-Based (Correlation)	Domain-Specific Validation
Akyash et al. (2025)	Controlled Experimental Study	Type III: Continuous Score Prediction	Direct Grader	Level 0: Fully Automated	Agreement-Based (Correlation)	Single-Course Validation
Grandel et al. (2025)	Controlled Experimental Study	Type IV: Rubric-Based Analytical Grading	Assistant to Human Grader	Level 2: Human-in-the-Loop	Agreement-Based (Accuracy)	Multi-Exam Validation
Wu-Ja et al. (2025)	Comparative Experimental Study	Type II: Discrete Category-Based Grading	Direct Grader	Level 0: Fully Automated	Agreement-Based (Accuracy)	Multi-Exam Validation

Overall, the corpus is dominated by controlled experimental studies and single-course validation settings, whereas large-scale comparative evidence remains comparatively limited. (Mohamed et al., 2025), (Mahdaoui et al., 2025), (Yousef et al., 2025).

half the corpus; Tables 1 and 3) is best explained not by pedagogical superiority but by methodological convenience and institutional compatibility.

Numeric instructor grades are routinely available and readily incorporated into quantitative evaluation pipelines, which makes continuous score prediction the path of least resistance for validating an LLM-based grader: it can be benchmarked directly against existing grading records using familiar metrics (MAE, correlation, equivalence testing). The paradigm is therefore selected, in part, because it fits the evaluation tools the field already possesses.

This convenience embeds an assumption that is rarely examined: that numeric proximity to an instructor's grade is an adequate proxy for grading quality. Yet identical numeric deviations carry very different pedagogical weight depending on where they fall—a one-point discrepancy is inconsequential mid-range but decisive at a pass/fail boundary or a grade threshold.

Because the dominant paradigm optimizes for a quantity (numeric agreement) that is only loosely coupled to educational consequence, reported performance metrics in much of the corpus are systematically disconnected from the decisions grades actually drive. The apparent maturity signalled by strong agreement figures may therefore overstate the field's pedagogical readiness.

5.2. *Operational configuration, not grading paradigm alone, drives reported performance*

Cross-study performance differences cannot be attributed to grading paradigm alone, because studies sharing a paradigm diverge sharply in the mechanism used to operationalize the model—direct, prompt-based, fine-tuned, hybrid (LLM + ML), or assistant-oriented.

Within continuous score prediction (Type III) alone, the corpus contains direct, prompt-based, and hybrid implementations (Mohamed et al., 2025), (Mahdaoui et al., 2025), (Korthals et al., 2026), each with distinct validation profiles and deployment trade-offs. Prompt--based grading offers flexibility but is sensitive to prompt formulation; fine-tuned grading (Manikani et al., 2025), (Yousef et al., 2025), improves local alignment at the cost of labeled data and contextual dependence; hybrid architectures (Mahdaoui et al., 2025), (Korthals et al., 2026), distribute evaluative responsibility across components, trading robustness for implementation complexity; assistant-oriented designs (Cisneros-González et al., 2025), (Grandel et al., 2025), embed pedagogical judgment within human oversight rather than replacing it.

Comparing studies by grading paradigm while ignoring operational configuration conflates distinct sources of variation. Valid cross-study interpretation requires controlling for configuration—a distinction the present taxonomy makes explicit and that future comparative work should preserve.

5.3. *Rubric-based grading is underexplored because pedagogical richness resists cheap automation*

The comparative scarcity of rubric-based analytical grading reflects a structural tension—pedagogical richness is harder to automate and harder to validate—rather than a lack of recognized value.

Rubric-based approaches decompose assessment into explicit criteria, offering interpretability and alignment with instructional intent. The corpus shows a notable association between rubric-based grading and human oversight (Cisneros-González et al., 2025), (Karsa & Goldschmidt, 2025), (Grandel et al., 2025), yet it also contains fully automated rubric-based implementations (Mazzone et al., 2025),

Table 2
Evaluation strategies and methodological assumptions in LLM-Based automated grading studies.

Study	What Is Evaluated	Primary Metric(s)	Implicit Assumption	Main Methodological Limitation
Cisneros-González et al. (2025)	Alignment between LLM-assisted grading and instructor rubric scores	MAE, R^2	Instructor grading is pedagogically valid and internally consistent	Single-course, single-rubric setting without cross-rubric validation
Mohamed et al. (2025)	Correctness of LLM-generated grades relative to human scores	Accuracy, ±1 Accuracy	Numeric proximity to teacher scores adequately reflects grading quality	Benchmark-style tasks with limited pedagogical interpretation
Mahdaoui et al. (2025)	Similarity between predicted and instructor-assigned grades	MAE, R^2, Accuracy	Human grades provide a reliable ground truth	Validation restricted to Python and a single dataset
Yousef et al. (2025)	Agreement between LLM-generated and teacher-assigned categories	Absolute Difference	Discrete grading categories sufficiently capture assessment quality	Limited scale and lack of pedagogical validation of category definitions
Mendonça et al. (2025)	Statistical equivalence between LLM and human grading	TOST Equivalence Testing	Statistical equivalence implies practical grading interchangeability	Single-exam context without longitudinal validation
Karsa and Goldschmidt (2025)	Correlation between rubric-based LLM scores and instructor assessments	Correlation	Higher agreement indicates superior grading quality	Sensitive to variability and inconsistency in human grading
Bernik, Radosevic, and Cep (2025)	Consistency between preliminary LLM grading and final instructor decisions	Correlation	Preliminary grading can reliably support instructional decision-making	Case-study design without controlled comparison
Manikani et al. (2025)	Correct classification of SQL query correctness	Accuracy	Binary correctness reflects meaningful learning outcomes	Domain-specific design limits broader generalizability
Jukiewicz (2026)	Alignment between LLM and instructor-assigned scores	Correlation, Agreement Measures	Instructor grading provides a stable evaluation reference	Limited to a single introductory programming course
Dosaru et al. (2025)	Accuracy of GenAI-assisted detection of design patterns	Accuracy, Reliability Measures	Design-pattern detection reflects broader code quality assessment	Small sample size within a single object-oriented programming course
Korthals et al. (2026)	Effectiveness of self-consistency and selective review strategies	Accuracy, Bias Reduction Measures	Human graders are reliable and internally consistent	Restricted to five assignments from one course
Ibáñez et al. (2025)	Alignment between multimodal LLM scores and expert consensus	Inter-Rater Reliability	Expert consensus functions as an objective grading standard	Limited to UML class-diagram assessment
Mazzone et al. (2025)	Consistency of grading across rubric configurations	Accuracy, Consistency Measures	Rubric structure determines scoring fairness and interpretability	Single-course validation setting
Pathak et al. (2025)	Effectiveness of question-specific rubric design	Spearman Correlation, Cohen's Kappa	Task-specific rubrics improve logical assessment fidelity	Validation limited to two datasets (DSA and OOP)
Akyash et al. (2025)	Alignment between LLM and expert human evaluations	Accuracy, Consistency Measures	Expert evaluation constitutes the gold standard	Limited sample size within a single course
Grandel et al. (2025)	Comparison between LLM assessments and human grading	Accuracy, Precision, Recall	Human graders provide the definitive evaluation baseline	Restricted to Java-based CS courses within one institution
Wu-Ja et al. (2025)	Comparative performance of multiple LLMs across assignment types	Accuracy, Consistency Measures	Predictive accuracy reflects pedagogical utility	Limited qualitative evaluation of feedback quality

Across the reviewed corpus, evaluation practices are predominantly agreement-based and rely heavily on human grading as the implicit ground truth, while cross-rubric validation, longitudinal evaluation, and broader pedagogical validation remain comparatively limited.

Table 3
Distribution of methodological choices across the reviewed corpus (n = 17).

Dimension	Category	Studies	Count	%
Grading Paradigm	Type I: Binary/Test-Oriented	Manikani et al. (2025)	1	5.9%
	Type II: Discrete Category-Based	(Yousef et al., 2025), (Dosaru et al., 2025), (Wu-Ja et al., 2025)	3	17.6%
	Type III: Continuous Score Prediction	(Mendonça et al., 2025), (Mohamed et al., 2025), (Mahdaoui et al., 2025), (Bernik, Radosevic, & Cep, 2025), (Jukiewicz, 2026), (Korthals et al., 2026), (Ibáñez et al., 2025), (Akyash et al., 2025)	8	47.1%
	Type IV: Rubric-Based Analytical Grading	(Cisneros-González et al., 2025), (Karsa & Goldschmidt, 2025), (Mazzone et al., 2025), (Pathak et al., 2025), (Grandel et al., 2025)	5	29.4%
Human Involvement	Level 0: Fully Automated	(Mendonça et al., 2025), (Manikani et al., 2025), (Mohamed et al., 2025), (Yousef et al., 2025), (Jukiewicz, 2026), (Ibáñez et al., 2025), (Mazzone et al., 2025), (Pathak et al., 2025), (Akyash et al., 2025), (Wu-Ja et al., 2025)	10	58.8%
	Level 1: Human-on-the-Loop	(Mahdaoui et al., 2025), (Korthals et al., 2026)	2	11.8%
	Level 2: Human-in-the-Loop	(Cisneros-González et al., 2025), (Karsa & Goldschmidt, 2025), (Bernik, Radosevic, & Cep, 2025), (Dosaru et al., 2025), (Grandel et al., 2025)	5	29.4%
Evaluation Strategy	Agreement-Based	(Cisneros-González et al., 2025), (Mahdaoui et al., 2025), (Yousef et al., 2025), (Karsa & Goldschmidt, 2025), (Bernik, Radosevic, & Cep, 2025), (Jukiewicz, 2026), (Dosaru et al., 2025), (Korthals et al., 2026), (Ibáñez et al., 2025), (Mazzone et al., 2025), (Pathak et al., 2025), (Akyash et al., 2025), (Grandel et al., 2025), (Wu-Ja et al., 2025)	14	82.4%
	Accuracy-Based	(Manikani et al., 2025), (Mohamed et al., 2025)	2	11.8%
	Equivalence Testing	Mendonça et al. (2025)	1	5.9%

(Pathak et al., 2025), demonstrating that human involvement is not intrinsic to the paradigm. What distinguishes rubric-based work is the added burden of rubric operationalization, criterion consistency, and implementation complexity—precisely the costs that the dominant continuous-score paradigm avoids.

The paradigm best aligned with pedagogical goals is also the one least served by the field's current measurement-driven incentives. This explains its underrepresentation and identifies scalable, interpretable, rubric-aware grading as the most pedagogically consequential open problem.

5.4. The feasibility of full automation tracks domain structure, not LLM capability

Where full automation succeeds, success is enabled by the constrained, well-specified structure of the assessment domain rather than

by the intrinsic capability of the LLM.

Fully automated configurations cluster in constrained domains—SQL evaluation, benchmark tasks, narrowly scoped exercises—where correctness can be checked against reference outputs and accuracy-based validation is straightforward. Manikani et al. (Manikani et al., 2025) report strong SQL-grading performance in exactly such a structured setting, whereas studies on open-ended tasks (Mohamed et al., 2025), (Yousef et al., 2025) and hybrid continuous prediction (Mahdaoui et al., 2025), face greater ambiguity and lower agreement.

High accuracy reported in constrained domains is therefore weak evidence about open-ended grading, and should not be extrapolated. The binding constraint on automation is the openness of the problem space, not model power—so progress on open-ended grading will not come from larger models alone.

5.5. Human involvement is a design variable, and triaging reframes what should Be evaluated

Human involvement in the corpus functions less as a transitional limitation than as a deliberate design choice, and its most consequential form—triaging—exposes a gap in current evaluation methodology.

Fully automated configurations cluster in scalability- and benchmark-oriented studies, while pedagogically oriented designs retain instructors. An intermediate pattern is the triaging configuration, in which the LLM handles routine cases and human review concentrates on ambiguous, disputed, or high-stakes submissions; Karsa and Goldschmidt (Karsa & Goldschmidt, 2025), exemplify this, with instructor effort directed to flagged cases rather than spread uniformly.

If the practical value of LLM grading lies in reallocating scarce expertise rather than maximizing score agreement, then score agreement is the wrong primary metric for triaging systems. Evaluation should additionally assess case-selection quality—whether the system correctly identifies submissions that require human review. This criterion is essentially absent from the current corpus and constitutes a concrete methodological gap.

5.6. Agreement metrics encode a conflation of machine-learning and educational validation norms

The near-universal reliance on agreement-based evaluation (Table 3) reflects the importation of machine-learning benchmarking conventions into a domain where they do not establish the relevant form of validity.

Agreement metrics treat instructor grading as an unquestioned reference standard and quantify proximity to it (Gao et al., 2024), (Mendonça et al., 2025), (Cisneros-González et al., 2025). But agreement does not establish grading validity, nor does it account for the documented variability and subjectivity of human grading itself (Gao et al., 2024). Educational measurement offers better-suited tools—chance-corrected inter-rater reliability (Cohen's Kappa, Krippendorff's Alpha) and decision-oriented criteria (pass/fail agreement, grade-boundary accuracy)—that are largely unused in the corpus.

A system minimizing mean absolute error while occasionally misclassifying students across critical thresholds may be less educationally useful than one that reliably places students in the correct performance category. Until evaluation practice incorporates decision-oriented and chance-corrected measures, reported performance will remain a measure of benchmarking convenience rather than pedagogical soundness.

5.7. Narrow validation confines the field to claims of feasibility, not robustness

The systematic confinement of validation to single courses, exams, languages, or institutions (Tables 1–3) means the field can currently support claims of feasibility but not of robustness.

High performance in one narrowly defined setting does not imply reliable behavior under different pedagogical structures, grading cultures, or programming domains. The scarcity of multi-course, multi-domain, and longitudinal validation is not a peripheral limitation but a ceiling on the kind of conclusion the literature can justify.

Claims about “the field” derived from context-bound studies should be read as conjectures pending broader validation. Cross-context and longitudinal evaluation is the precondition for any transition from proof-of-concept to dependable practice.

5.8. Summary of insights

Taken together, these subsections support a single explanatory claim: the field's dominant design and evaluation choices are shaped largely by what is convenient to measure and operationalize, and this orientation systematically constrains the conclusions the literature can support. Continuous score prediction, full automation in constrained domains, and agreement-based validation each represent a path of least methodological resistance rather than a pedagogically optimal choice. Recognizing this is what distinguishes the parallel design trajectories in the corpus from genuine convergence, and it reframes the field's central challenge: not improving grading accuracy, but building evaluation and design frameworks whose measures track educational consequence rather than measurement convenience.

6. Research gaps and future directions

The comparative analysis identifies several systematic research gaps that currently constrain the maturity, interpretability, and broader applicability of LLM-based automated grading in programming education. These limitations arise not only from technical constraints but also from unresolved methodological and pedagogical challenges.

6.1. Conceptual ambiguity in automated grading

A first gap concerns the absence of a shared conceptualization of automated grading. As shown throughout the comparative analysis, reviewed studies operationalize grading in markedly different ways, ranging from binary correctness detection and continuous score prediction to rubric-based assessment and hybrid human–AI workflows. While this diversity reflects an emerging field, it complicates cross-study comparison and limits cumulative knowledge building.

Future research would benefit from clearer alignment between grading paradigms and educational objectives, explicitly distinguishing whether automated grading is intended to approximate instructor scores, support formative feedback, triage human review, or automate specific components of the assessment process.

6.2. Beyond agreement metrics: toward educationally grounded validation

Most reviewed studies evaluate performance primarily through agreement-based metrics that quantify proximity to human-assigned grades. This strategy implicitly treats instructor grading as a stable and pedagogically valid reference standard, an assumption that is rarely examined explicitly.

Future work should broaden evaluation frameworks beyond numeric agreement alone by incorporating educational measurement approaches, including inter-rater reliability analysis, decision-oriented validation (e.g., pass/fail consistency or grade-boundary accuracy), and robustness to variability in human grading practices.

6.3. Expanding rubric-based and pedagogically structured grading

Although rubric-based analytical grading offers strong pedagogical alignment and greater criterion transparency, it remains comparatively underrepresented within the reviewed corpus. While several rubric-driven studies rely on substantial human oversight, the corpus also

includes emerging fully automated implementations, suggesting an evolving but still methodologically immature research area.

Future research should investigate scalable, interpretable, and rubric-aware grading mechanisms capable of preserving pedagogical structure while balancing automation, consistency, and instructional accountability.

6.4. Generalization beyond local educational contexts

A consistent limitation across the reviewed studies concerns restricted validation scope. Most evaluations remain confined to single courses, examinations, programming languages, or institutional settings, limiting confidence in the robustness and transferability of reported findings.

Future work should prioritize broader validation across curricula, learner populations, programming domains, and longitudinal educational settings to determine whether current systems capture general grading capability or context-specific optimization.

6.5. Human oversight as a design variable

Human participation is frequently framed as a transitional limitation to be minimized. However, the comparative findings suggest that human involvement often functions as a deliberate design component, particularly in pedagogically sensitive or high-stakes assessment contexts.

Future research should model human oversight more explicitly as a configurable design dimension, investigating selective review mechanisms, triaging workflows, and collaborative human–AI assessment configurations alongside traditional fully automated paradigms.

6.6. Toward mature and pedagogically grounded grading systems

Moving the field beyond proof-of-concept implementations will likely require a shift from performance-centric benchmarking toward pedagogically grounded system design and validation. This transition includes clearer specification of grading objectives, diversified evaluation methodologies, broader validation contexts, and stronger integration of educational assessment principles into system development.

Additional research directions—including explainable grading, fairness-aware evaluation, multimodal assessment designs, institutional deployment strategies, and computational sustainability—may further support the development of reliable and educationally meaningful LLM-based grading systems.

7. Conclusion

This study presented a comparative methodological analysis of recent research on LLM-based automated grading in programming education. Based on a systematically selected corpus of 17 peer-reviewed studies published between 2023 and 2026, including journal articles and computing education conference publications, the study developed a taxonomy encompassing grading paradigms, LLM operational roles, human involvement, evaluation strategies, generalization scope, and levels of evidence.

The findings indicate that variation in reported performance across studies is strongly associated with methodological heterogeneity rather than with directly contradictory results. The reviewed literature is characterized by diverse grading paradigms, heterogeneous validation practices, differing configurations of human oversight, and predominantly context-specific evaluation settings. Continuous score prediction remains the most frequently adopted paradigm, whereas rubric-based approaches, despite their pedagogical relevance, remain comparatively less represented and unevenly explored.

The comparative analysis further highlights several methodological limitations that currently constrain the interpretability and broader applicability of reported findings. Most studies rely on agreement-based evaluation metrics, assume instructor grading as a stable reference standard, and conduct validation within single-course or domain-specific contexts. In addition, the role of human involvement varies substantially across studies, reflecting differing assumptions regarding automation, accountability, and pedagogical alignment.

Rather than supporting a single preferred grading configuration, the findings emphasize the importance of aligning methodological design choices with explicit educational objectives and assessment requirements. Advancing the field will likely require clearer conceptual definitions of automated grading, broader validation across educational contexts, more diversified evaluation frameworks, and more systematic treatment of human oversight within grading system design.

From a practical perspective, the findings suggest that fully automated configurations may be more suitable for constrained and well-defined assessment scenarios, whereas human-AI collaborative approaches may be better aligned with contexts requiring interpretability, pedagogical nuance, or evaluative accountability.

Overall, the proposed taxonomy and comparative synthesis provide a structured basis for interpreting existing work and may support the development of more robust, transparent, and pedagogically grounded automated grading systems in programming education.

8. Limitations of the study

Several limitations should be considered when interpreting the findings of this study.

First, although the study examines a recent corpus of peer-reviewed literature on LLM-based automated grading in programming education, the field remains rapidly evolving. New models, evaluation practices, and application contexts continue to emerge, meaning that the methodological patterns identified in this analysis may evolve as additional research becomes available.

Second, the review relies on searches conducted through Scopus and Web of Science. Although these databases provide broad coverage of high-quality peer-reviewed venues, including a substantial portion of ACM- and IEEE-published research, relevant studies indexed only in specialized digital libraries or venue-specific proceedings repositories—such as those associated with AIED, EDM, or L@S—may not have been fully captured. Consequently, the findings remain partially dependent on database coverage and search-strategy design.

Third, the analysis is based on the methodologies, evaluation practices, and reporting detail available in the selected publications. Because the study adopts a secondary analytical perspective, methodological assumptions and limitations were interpreted from published descriptions rather than independently verified through replication or empirical testing. The resulting interpretations should therefore be understood as comparative analytical observations rather than definitive judgments regarding individual studies.

Finally, this study does not evaluate the empirical performance of specific LLMs, prompts, or grading systems. Instead, it adopts a methodological and comparative perspective focused on design choices, validation strategies, and research structure. Accordingly, the conclusions should be interpreted as insights into the organization, assumptions, and current state of the research landscape rather than as direct claims regarding the superiority or effectiveness of particular grading approaches.

CRediT authorship contribution statement

Abderazzak EL Mernissi: Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Fatima Lakrami:** Writing – review & editing, Validation, Supervision, Methodology. **Hicham Toumi:** Writing – review & editing, Validation, Supervision, Methodology.

Ethical statement

This study did not involve human participants, human data, or human tissue. The research is based exclusively on the analysis of published literature. Therefore, ethical approval and informed consent were not required.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Akyash, M., Azar, K. Z., & Kamali, H. M. (2025). StepGrade: Grading programming assignments with context-aware LLMs. In *IEEE Integr. STEM Educ. Conf., ISEC*. Institute of Electrical and Electronics Engineers Inc.. https://doi.org/10.1109/ISEC64801.2025.11147374

Ala-Mutka, K. M. (2005). A survey of automated assessment approaches for programming assignments. *Computer Science Education, 15*(2), 83–102. https://doi.org/10.1080/08993400500150747

Bernik, A., Radošević, D., & Čep, A. (2025). A comparative study of large language models in programming education: Accuracy, efficiency, and feedback in student assignment grading. *Appl. Sci., 15*(18). https://doi.org/10.3390/app151810055

Bernik, A., Radosevic, D., & Cep, A. (2025). AI-Powered learning: Revolutionizing education and automated code evaluation. *Information, 16*(11), 1015. https://doi.org/10.3390/info16111015

Cheang, B., Kurnia, A., Lim, A., & Oon, W.-C. (2003). On automated grading of programming assignments in an academic institution. *Computer Education, 41*(2), 121–131. https://doi.org/10.1016/S0360-1315(03)00030-7

Cisneros-González, J., Gordo-Herrera, N., Barcia-Santos, I., & Sánchez-Soriano, J. (2025). JorGPT: Instructor-aided grading of programming assignments with large language models (LLMs). *Future Internet, 17*(6). https://doi.org/10.3390/fi17060265

Dong, Y., Hou, J., & Lu, X. (2020). An intelligent online judge system for programming training. In *Database systems for advanced applications: 25th international conference, DASFAA 2020, Jeju, South Korea, September 24–27, 2020, proceedings, part III* (pp. 785–789). Berlin, Heidelberg: Springer-Verlag. https://doi.org/10.1007/978-3-030-59419-0_57.

Dosaru, D.-F., Simion, D.-M., Ignat, A.-H., Negreanu, L.-C., & Olteanu, A.-C. (2025). Using GenAI to assess design patterns in student-written code. *IEEE Transactions on Learning Technologies, 18*, 869–876. https://doi.org/10.1109/TLT.2025.3604054

Douce, C., Livingstone, D., & Orwell, J. (2005). Automatic test-based assessment of programming: A review. *J. Educ. Resour. Comput. JERIC, 5*(3), 4–es. https://doi.org/10.1145/1163405.1163409

A. Efremov, A. Ghosh, and A. Singla, "Zero-shot Learning of Hint Policy via Reinforcement Learning and Program Synthesis".

Feng, Z., et al. (2020). CodeBERT: A pre-trained model for programming and natural languages. *arXiv: arXiv:2002.08155*. https://doi.org/10.48550/arXiv.2002.08155

Gao, R., Merzdorf, H. E., Anwar, S., Hipwell, M. C., & Srinivasa, A. R. (2024). Automatic assessment of text-based responses in post-secondary education: A systematic review. *Computers and Education: Artificial Intelligence, 6*, Article 100206. https://doi.org/10.1016/j.caeai.2024.100206

Grandel, S., Schmidt, D. C., & Leach, K. (2025). Applying large language models to enhance the assessment of java programming assignments. *Presented at the Proceedings of the ACM SIGSOFT Symposium on the Foundations of Software Engineering*, 789–799. https://doi.org/10.1145/3696630.3727236

Guo, D., et al. (2021). GraphCodeBERT: Pre-training code representations with data flow. *arXiv: arXiv:2009.08366*. https://doi.org/10.48550/arXiv.2009.08366

Hollingsworth, J. (1960). Automatic graders for programming classes. *Communications of the ACM, 3*(10), 528–529. https://doi.org/10.1145/367415.367422

Ibáñez, M. B., Barrón-Estrada, M. L., & Zatarain-Cabada, R. (2025). Can multimodal large language models grade like an expert? A study on UML class diagram assessment accuracy. *Computer Applications in Engineering Education, 33*(5). https://doi.org/10.1002/cae.70080

Jiang, E., & Fowler, M. (2026). You don't need a data center to explain in plain English! comparing open-source and propriety LLMs for EiPE grading. *Presented at the SIGCSE TS 2026 - Proceedings of the 57th ACM Technical Symposium on Computer Science Education, 1*, 512–518. https://doi.org/10.1145/3770762.3772560

Jukiewicz, M. (2024). The future of grading programming assignments in education: The role of ChatGPT in automating the assessment and feedback process. *Thinking Skills and Creativity, 52*, Article 101522. https://doi.org/10.1016/j.tsc.2024.101522

Jukiewicz, M. (2026). A systematic comparison of large language models for automated assignment assessment in programming education: Exploring the importance of architecture and vendor. *Computers and Education Open, 10*, Article 100364. https://doi.org/10.1016/j.caeo.2026.100364

Karsa, Z. I., & Goldschmidt, B. (2025). Automatic evaluation of programming tasks supported by language models. *IEEE Access, 13*, 147741–147756. https://doi.org/10.1109/ACCESS.2025.3601448

Korthals, L., Akrong, E., Geller, G., Rosenbusch, H., Grasman, R., & Visser, I. (2026). Towards reliable LLM grading through self-consistency and selective human review: Higher accuracy, less work. *Mach. Learn. Knowl. Extr., 8*(3), 74. https://doi.org/10.3390/make8030074

Mahdaoui, M., Nouh, S., El Kasmi Alaoui, M. S., & Kandali, K. (2025). Automated grading method of python code submissions using large language models and machine learning. *Inf. Switz., 16*(8). https://doi.org/10.3390/info16080674

Manikani, K., Chapaneri, R., Shetty, D., & Shah, D. (2025). SQL autograder: Web-based LLM-Powered autograder for assessment of SQL queries. *International Journal of Artificial Intelligence in Education, 35*(4), 2047–2077. https://doi.org/10.1007/s40593-025-00460-2

Mazzone, S. B., Forden, J., & Brylow, D. (2025). Can LLMs be fair graders? Only if you bribe them with the right rubric. *Presented at the proceedings - Frontiers in education conference, FIE*. https://doi.org/10.1109/FIE63693.2025.11328653

Mendonça, P. C., Quintal, F., & Mendonça, F. (2025). Evaluating LLMs for automated scoring in formative assessments. *Appl. Sci. Switz., 15*(5). https://doi.org/10.3390/app15052787

Messer, M., Brown, N. C. C., Kölling, M., & Shi, M. (2024). Automated grading and feedback tools for programming education: A systematic review. *ACM Transactions on Computing Education, 24*(1), 1–43. https://doi.org/10.1145/3636515

Mohamed, K., Yousef, M., Medhat, W., Mohamed, E. H., Khoriba, G., & Arafa, T. (2025). Hands-on analysis of using large language models for the auto evaluation of programming assignments. *Information Systems, 128*, Article 102473. https://doi.org/10.1016/j.is.2024.102473

Moorhouse, B. L., Yeo, M. A., & Wan, Y. (2023). Generative AI tools and assessment: Guidelines of the world's top-ranking universities. *Computers and Education Open, 5*, Article 100151. https://doi.org/10.1016/j.caeo.2023.100151

Orr, J. W., & Russell, N. (2021). Automatic assessment of the design quality of python programs with personalized feedback. *arXiv: arXiv:2106.01399*. https://doi.org/10.48550/arXiv.2106.01399

Paiva, J. C., Leal, J. P., & Figueira, Á. (2022). Automated assessment in computer science education: A state-of-the-art review. *ACM Trans. Comput. Educ. TOCE, 22*(3), 34: 1–34:40. https://doi.org/10.1145/3513140

Pathak, A., et al. (2025). Rubric is all you need: Improving LLM-based code evaluation with question-specific rubrics. *Presented at the PROCEEDINGS of the ACM CONFERENCE on INTERNATIONAL COMPUTING EDUCATION RESEARCH, 1*, 181–195. https://doi.org/10.1145/3702652.3744220. ICER 2025.

Pereira, A. F., & Ferreira Mello, R. (2025). A systematic literature review on large language models applications in computer programming teaching evaluation process. *IEEE Access, 13*, 113449–113460. https://doi.org/10.1109/ACCESS.2025.3584060

Pwanedo Amos, J., et al. (2025). A bibliometric exposition and review on leveraging LLMs for programming education. *IEEE Access, 13*, 58364–58393. https://doi.org/10.1109/ACCESS.2025.3554627

Souza, D. M., Felizardo, K. R., & Barbosa, E. F. (2016). A systematic literature review of assessment tools for programming assignments. In *2016 IEEE 29th international conference on software engineering education and training (CSEET)* (pp. 147–156). https://doi.org/10.1109/CSEET.2016.48

Ullah, Z., Lajis, A., Jamjoom, M., Altalhi, A., Al-Ghamdi, A., & Saleem, F. (2018). The effect of automatic assessment on novice programming: Strengths and limitations of existing systems. *Computer Applications in Engineering Education, 26*(6), 2328–2341. https://doi.org/10.1002/cae.21974

Weißenrieder, F., Lenzer, S., & Mühling, A. (2025). Assessing computer science pedagogical content knowledge: A systematic literature review. *Computer Science Education, 0*(0), 1–42. https://doi.org/10.1080/08993408.2025.2586872

Wu-Ja, L., Hui-Lin, W., & Yu-Guang, Z. (2025). A comparative study of automated grading on programming assignments based on LLMs. In *2025 IEEE international conference on consumer electronics - Taiwan (ICCE-Taiwan)* (pp. 429–430). https://doi.org/10.1109/ICCE-Taiwan66881.2025.11208057

Yousef, M., Mohamed, K., Medhat, W., Mohamed, E. H., Khoriba, G., & Arafa, T. (2025). BeGrading: Large language models for enhanced feedback in programming education. *Neural Computing & Applications, 37*(2), 1027–1040. https://doi.org/10.1007/s00521-024-10449-y

Zampirolli, F. de A., Teubl, F., Pisani, P. H., & e Silva, T. A. P. (2026). Intelligent feedback for individualized introductory programming exercises. *Computer Applications in Engineering Education, 34*(1), Article e70132. https://doi.org/10.1002/cae.70132